

VR Games Based on Circumplex Model of Group Tasks for QoE Measurements and Beyond

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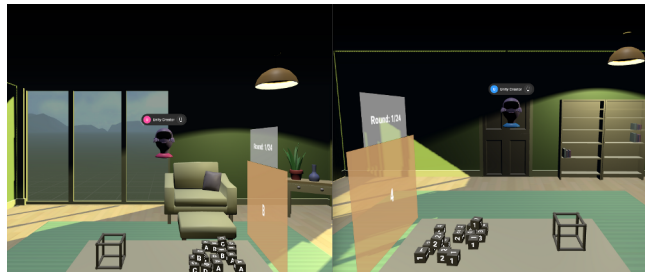


Figure 1: Screenshot from our demo, in this picture users are playing Cognitive Conflict task

ABSTRACT

This research demo presents a system for studying collaboration in Virtual Reality. The system was developed in Unity and provides a multi-user VR simulator with a set of collaborative tasks categorized according to McGrath's circumplex model of group tasks. The platform is fully configurable, allowing researchers to adapt tasks and parameters to their own studies on VR cooperation. The system integrates in-VR subjective assessment, enabling users to rate Quality of Experience without removing the headset. In addition, it automatically collects performance data to support quantitative analysis. In this demo, participants collaborate in dyads while our system varies the network delay using the Clumsy tool. The demo presents a novel approach to studying subjective QoE in networked VR cooperation.

Index Terms: Human-centered-computing, Quality of Experience, Virtual Reality, Network Delay.

1 INTRODUCTION

The growing advancement of virtual reality leads to the increasing number of applications that allow meetings and cooperation in virtual worlds [2]. Moreover, one of the assumed directions in the evolution of Internet-delivered experiences is the growing prominence of 3D immersive content [3]. Preparation for the wider adoption of Extended Reality (XR) solutions covers research analyzing the various factors impacting Quality of Experience in XR like: network [10, 12], interaction in XR [4, 9], cybersickness [5, 8], immersion [7], and user collaboration in XR [11, 13].

An especially important factor influencing quality for XR communication is the purpose people communicate. Depending on the purpose, the quality can be different even if all other factors are the same. In psychology, the complex group interaction was systematically classified by McGrath circumplex model of group tasks [6] (see Figure 2). This model integrated key insights from earlier frameworks, delivering a model divided into four main quadrants: Generate, Choose, Negotiate, and Execute. Each quadrant contains two categories of tasks, resulting in a total number of 8

categories. We decided to use this theory in our system to provide a wide spectrum of tasks, allowing researchers to perform a comprehensive analysis of subjective and objective metrics of VR collaboration.

We used Unity to prepare the VR environments. To ensure the multiplayer feature, we used the Unity VR Multiplayer template. Specific games were designed and developed by ourselves.

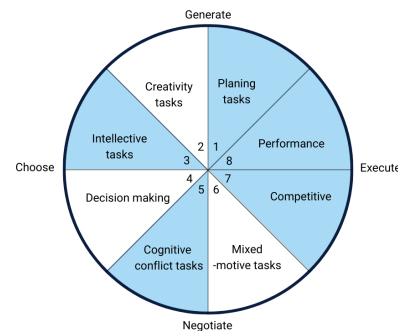


Figure 2: A circular diagram showing the circumplex model of group tasks. With color, we noted which tasks are available to play in our demo. Adapted from [6]

2 SYSTEM

In line with ITU-T recommendation P.1305, a testing system should allow for repeated interactions that are similar but not identical. Similarity allows for a single explanation for a series of experiments. At the same time, each game is slightly different, so we can analyze results independently. With the same games, playing it for the first or the second time would influence results significantly.

In order to achieve that goal, we developed a series of games. All games are based on the similar concept of moving cubes with additional information displayed on a virtual screen. Each game is short and contains a random element, allowing for repetitions.

2.1 VR environment

Our system is a VR simulator prepared in Unity engine (See Fig. 1). The games we prepared cover 5 out of 8 categories described by McGrath: Intellectual, Cognitive Conflict, Planning, Performance,

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and Competitive. Below, details of each game are described. The demo allows you to play any of this game with varying network conditions. **Intellectual:** This task requires participants to decode a short word using a monoalphabetic substitution cipher. One participant sees the encoded word, while the other has access to the substitution rules; only the participant viewing the encoded word can place cubes. Successful completion depends on verbal communication and shared reasoning. **Cognitive Conflict:** Sixteen cubes are placed on a table, but each participant perceives different information: one sees letters, the other sees numbers. Participants have to find the cube with the correct number and letter. Since both letters and numbers are repeated, they have to communicate to find the only one correct cube. **Planning:** This task extends the cognitive conflict scenario by introducing a limit on the number of allowed cube interactions. Exceeding this limit causes all cubes to reset, encouraging participants to plan actions in advance and coordinate verbally. The interaction limit is configurable. **Performance:** In this variant, task difficulty is defined by a time constraint for placing the correct cube. If the time limit is exceeded, the cubes reset and visual feedback is provided. Each failure increases the time before the next reset. This ensures pressure on performance while maintaining task solvability. **Competitive:** Participants independently solve separate cube-selection tasks within a shared environment. Each player inserts cubes into their box, receiving immediate feedback, while used cubes respawn to prevent deadlock. Although communication is unnecessary, shared spatial interaction may lead to interference between players.

All of these tasks can be freely configured up to the researcher's idea, which makes our solution very flexible in subjective VR cooperation studies.

2.2 Data collection

To ensure the immersive experience during the research we decided to ask for users perceived QoE directly inside the VR simulation. After each session there is a board with Absolute Category Rating (see ITU-T P.910) asking about interaction quality on a 1 to 5 scale. Users answer by pressing the suitable button on the board; the answer is then saved to a CSV file for further analysis.

Our system also gathers performance data by measuring the time of different actions. Touching a cube, placing correct cube, placing wrong cube, and reroll cubes. This data is stored in a CSV file. The recorded times can be used to calculate further metrics like: time to solve a puzzle, number of errors before solving, or number of interactions before solving.

3 NETWORK DISTORTION

Our system uses a Windows server for between players communication. At the server a **Clumsy** [1] tool allows for distorting the network by introducing delay, packet loss, and bandwidth limitation. Since each interaction goes through the server, we are sure that specific conditions influence communication. Changing the network condition allows for testing different conditions.

4 DEMO

In this demo users can play together in a shared virtual space in any of our five games. We are distorting the interaction using the Clumsy tool. Nevertheless, the audio is not distorted since the practicality of the demo makes it very hard to limit direct communication. The players are close to each other. Our experience shows that it is advantageous to have no delay on audio and delayed interaction since the second is more visible in such cases.

At the beginning, two users are instructed about how to play the game (movement and interaction), and they are also informed what the task is in the specific game. We are going to show two extreme network conditions so the players can quickly understand how the interaction changes depending on the delay. If we have more time,

we can show a full set of different network conditions as used in our laboratory.

After finishing the game, we show the users how they rated the quality for various conditions, and also we can show them an analysis of their performance during the game.

5 APPLICATIONS

This system can be used in various studies about the network's impact on the QoE of users collaboration. It can be freely configured up to the researcher's idea. In our research we introduced a network delay between users, but it is also possible to analyze other conditions like jitter or packet loss. Our system provides different tasks based on psychological theory, which adds the important value of studying how people feel about the network issues during different activities. This system can also be used for other studies, like cooperating with AI agents or analyzing how the environment impacts the work quality for users.

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